Modeling and Performance Analysis of a Laval Nozzle using StarCCM+

TP 2 - Computational Fluid Dynamics (CFD) Lab Report

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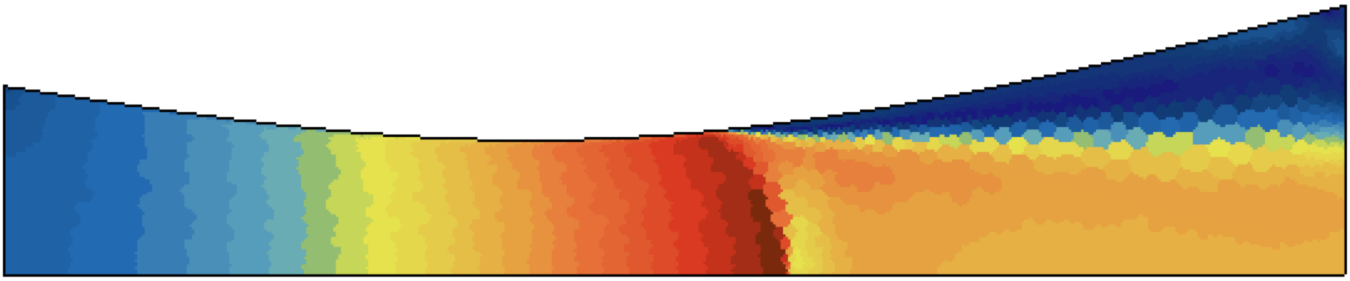
ABSTRACT

This report investigates the aerodynamic behavior of a convergent-divergent (Laval) nozzle under compressible flow conditions. Using StarCCM+, a 2D axisymmetric model was developed to analyze the conversion of thermal energy into kinetic energy through cross-sectional variations. The study includes an analytical validation of exit pressure and mass flow rates, followed by a numerical mesh convergence test. The influence of different outlet pressure conditions (P_s = 300 kPa and P_s = 500 kPa) is evaluated to understand nozzle performance and flow regimes.

Key words. Laval Nozzle — Compressible Flow — StarCCM+ — Isentropic Flow — Mach Number



1. Introduction

Laval nozzles are critical components in jet engines and propulsion systems, designed to accelerate gases to supersonic velocities. By exploiting the relationship between pressure reduction and velocity increase through varying cross-sections, these nozzles convert internal thermal energy into kinetic thrust.

This practical work focuses on the CFD analysis of a nozzle transporting air. The objective is to study compressible flow and analyze nozzle performance as a function of various operating conditions.

2. Analytical Study

2.1. Fluid Properties

The air is modeled as a perfect gas with the following properties:

- **Adiabatic index (γ):** 1.4
- **Perfect gas constant (R):** 287 J/(kg·K)
- **Heat capacity (C_p):** 1005 J/(kg·K)

The inlet (stagnation) conditions are as follows:

- **Stagnation pressure (P_0):** 1000 kPa
- **Stagnation temperature (T_0):** 500 K

2.2. Geometry

The nozzle geometric parameters are defined by:

- **Critical area (A^*):** 0.002 m^2
- **Outlet area (A_s):** 0.008 m^2

2.3. Theoretical Calculations

For the "design" condition with a Mach number of $M = 2.94$ at the exit, the exit pressure P_s is calculated as:

$$P_s = P_0\left(1 + \frac{\gamma - 1}{2}M^2\right)^{-\frac{\gamma}{\gamma-1}} = 29795.0138Pa$$

The theoretical maximum mass flow rate is given by the following formula:

$$\dot{m} = P_0 A^* \sqrt{\frac{\gamma}{RT_0}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}}$$

We can see that the value is indeed around $3.61kg/s$.

3. Numerical Study (StarCCM+)

3.1. Modeling

Since the flow is axisymmetric, the modeling is carried out in a 2D axisymmetric configuration. The geometry was extracted from a 3D extrusion with an arbitrary thickness of 1 cm. After following the given dimensions, we obtain the 3D model :

Fig. 1. Nozzle geometry

Then we name all the faces as follows:

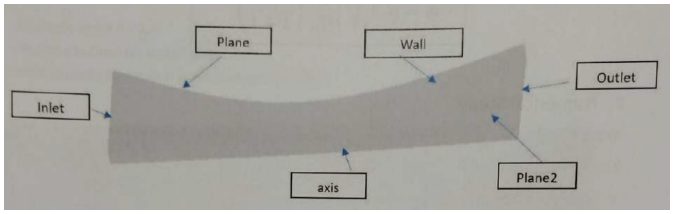


Fig. 2. Named faces

3.2. Mesh Generation and Refinement

The discretization of the computational domain is a critical step in ensuring the convergence and accuracy of the supersonic flow simulation. In this study, the mesh was generated using an automated 2D approach within the Star-CCM+ framework to adapt the extruded 3D geometry into a 2D axisymmetric plane.

3.2.1. Meshing Models

The mesh was created using the **Automated Mesh (2D)** operation. To correctly capture the complex physics of the compressible flow within the nozzle, two complementary meshing models were utilized:

- **Polygonal Mesher:** This model was chosen for the core flow discretization. Polygonal cells offer superior numerical stability and reduced numerical diffusion compared to tetrahedral meshes, which is particularly beneficial for resolving shock waves and high-gradient regions in the divergent section.
- **Prism Layer Mesher:** To accurately resolve the viscous effects and the velocity gradients near the nozzle walls, a prism layer mesh was implemented. This ensures that the boundary layer development is correctly captured, which is vital for calculating potential flow separation during over-expanded regimes.

3.2.2. Grid Parameters and Boundary Layer Control

The mesh density was controlled through a set of global parameters defined in the **Default Controls**. The following values were applied for the baseline simulation:

- **Base Size:** Set to 0.005 m to provide a consistent reference scale for the entire domain.
- **Number of Prism Layers:** 5 layers were generated to discretize the near-wall region.
- **Prism Layer Stretching:** A factor of 1.3 was used to provide a smooth geometric transition from the wall towards the core mesh.
- **Prism Layer Total Thickness:** Defined as 30% of the local mesh size to ensure sufficient coverage of the hydrodynamic boundary layer.

The resulting mesh represents a 2D section of the nozzle, identified in the software by a red dot in the Parts tree, confirming that the geometry is correctly prepared for the 2D axisymmetric solver.

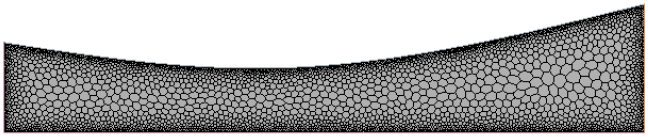


Fig. 3. Mesh

4. Numerical Results

4.1. Mesh Convergence Table

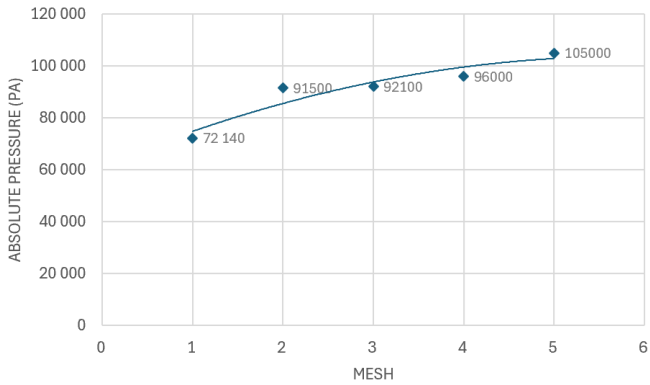


Fig. 4. Mesh convergence

4.2. Physics Models

The selected physical models are:

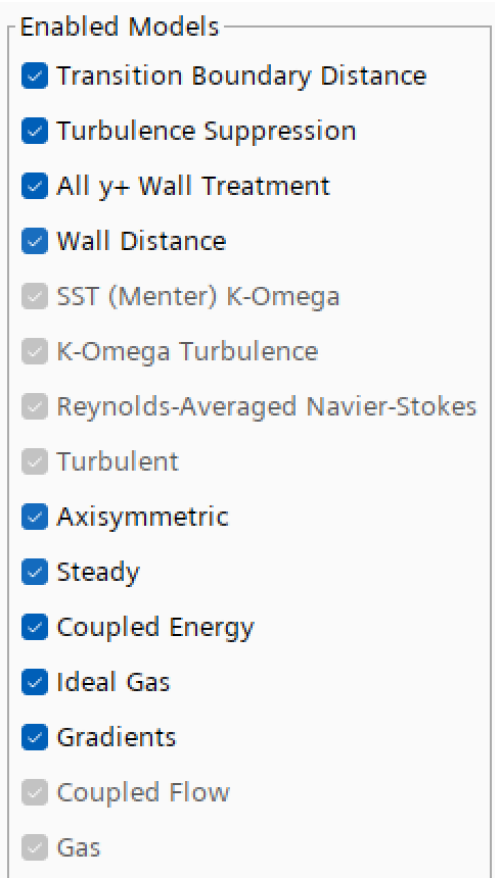


Fig. 5. Physics model

After creating scalar scenes and XY plots, we can observe for a base size of 0.6mm.

4.3. Scalar scenes and XY Plots (for different outlet pressure)

4.3.1. **Scalar scenes and XY Plots (for initial pressure)**

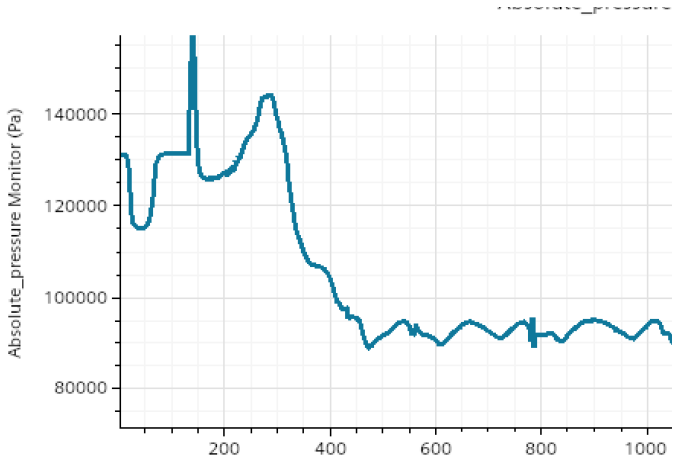


Fig. 6. Absolute pressure

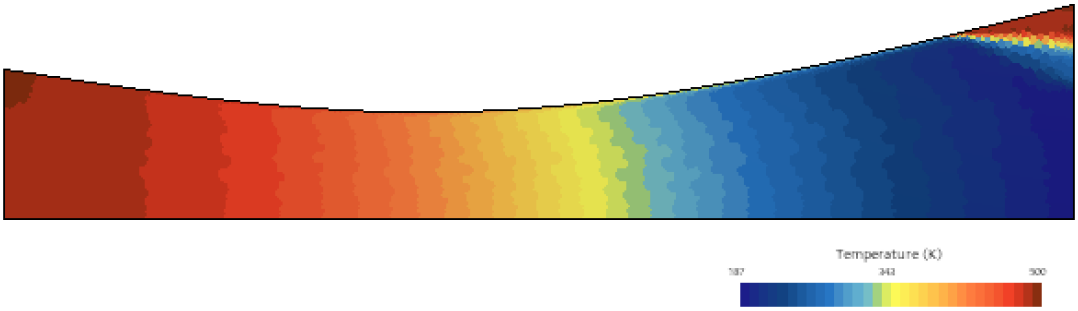


Fig. 7. Temperature

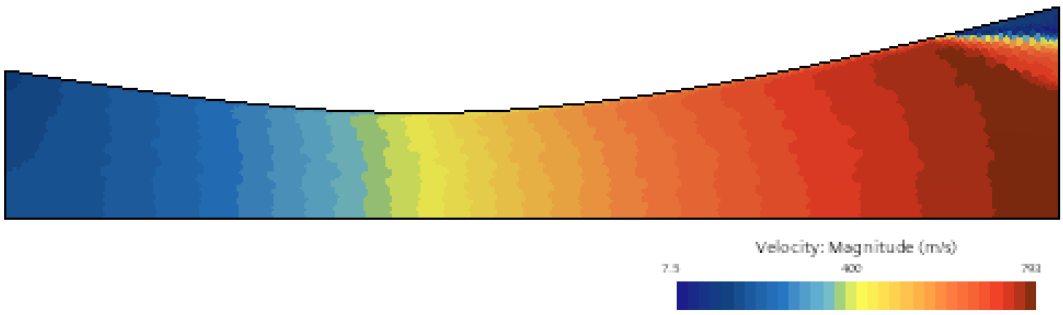


Fig. 8. Velocity magnitude

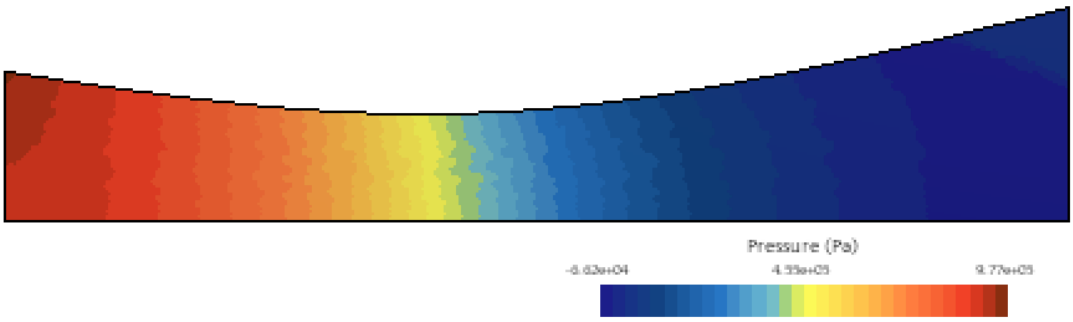


Fig. 9. Pressure

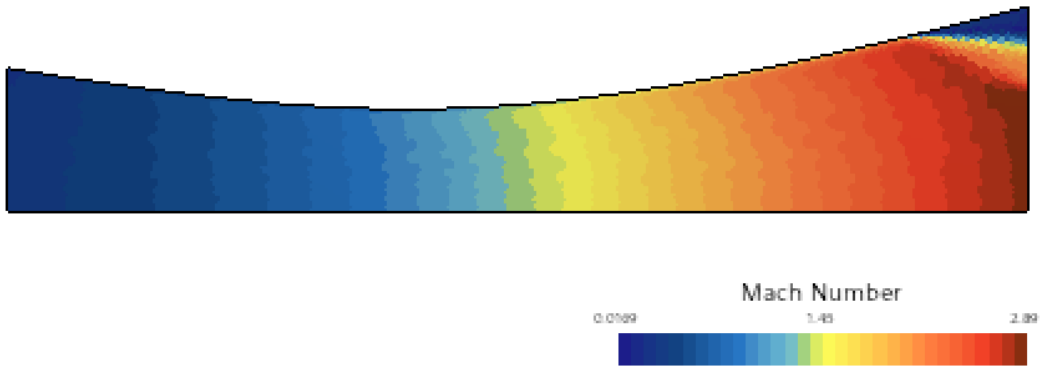


Fig. 10. Mach number

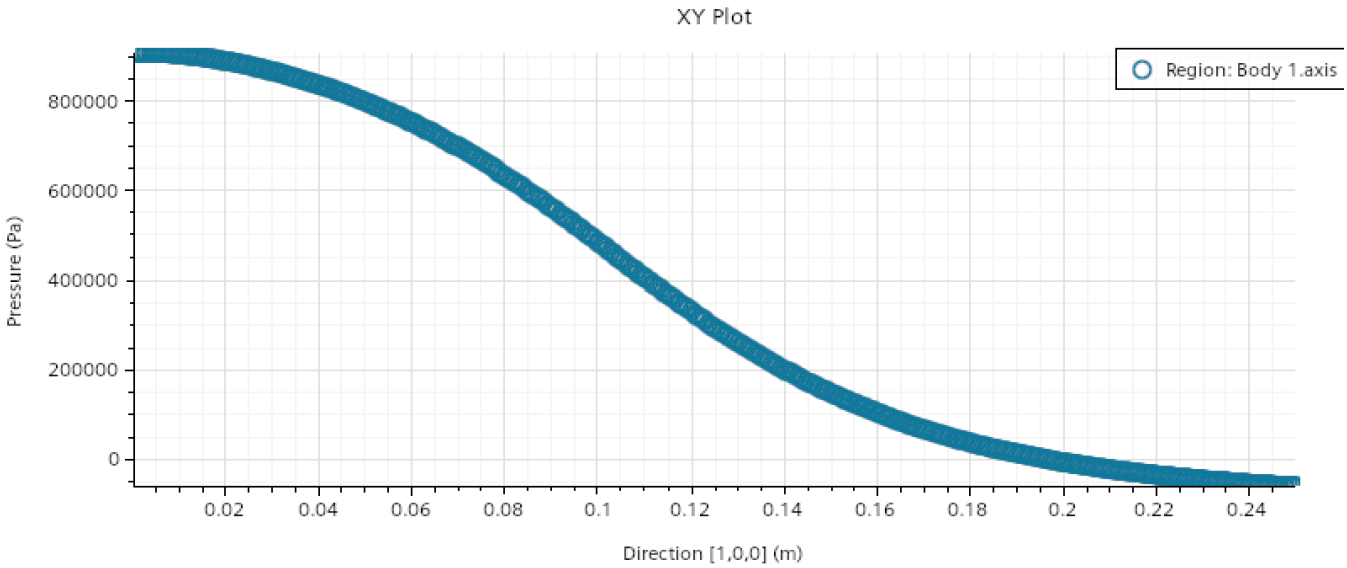


Fig. 11. XY Plot : Pressure

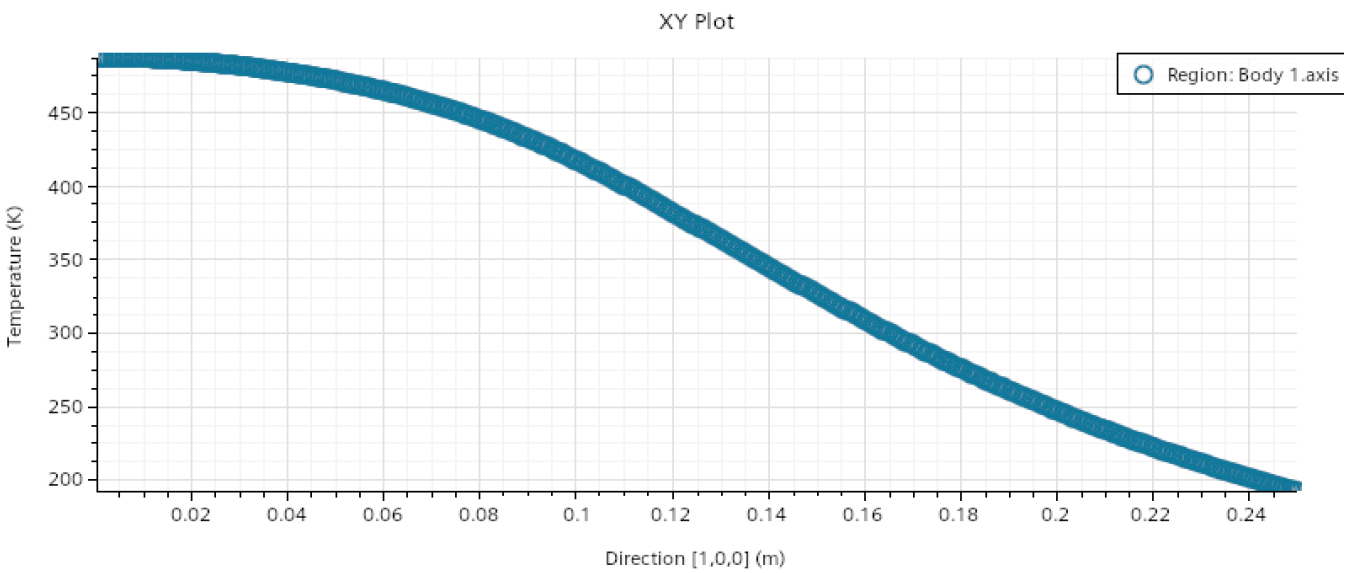


Fig. 12. XY Plot : Temperature

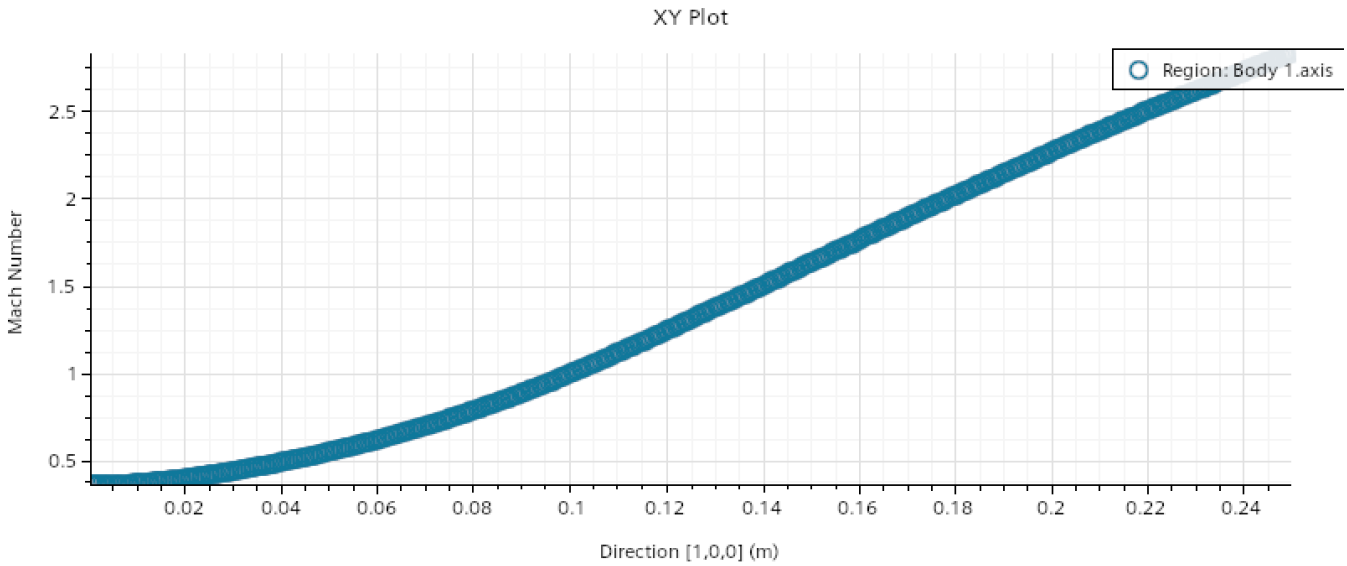


Fig. 13. XY Plot : Mach Number

4.3.2. **Scalars and XY Plots (for 300kPa)**

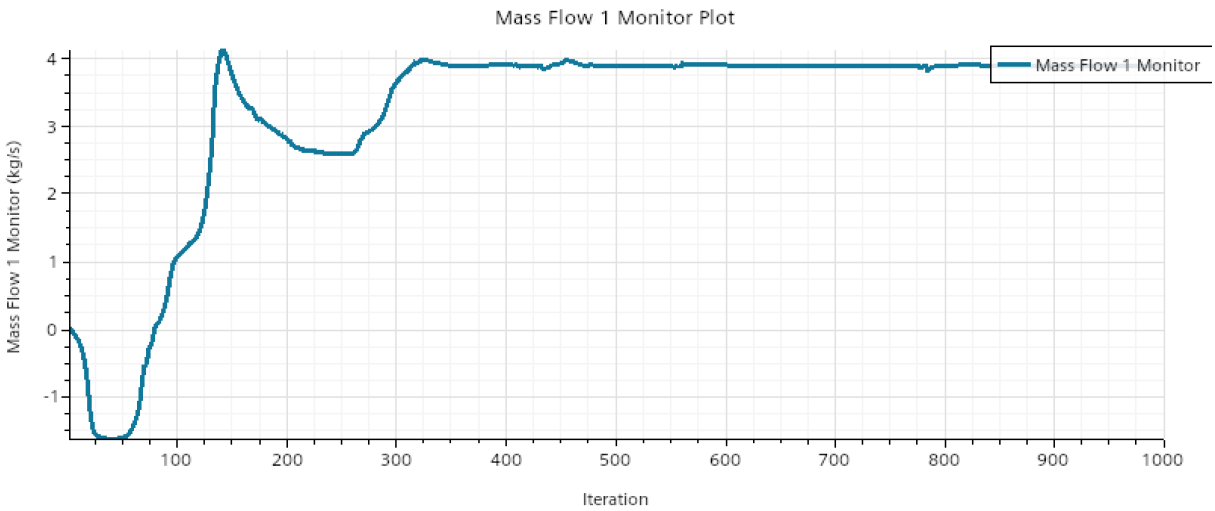


Fig. 14. Mass flow

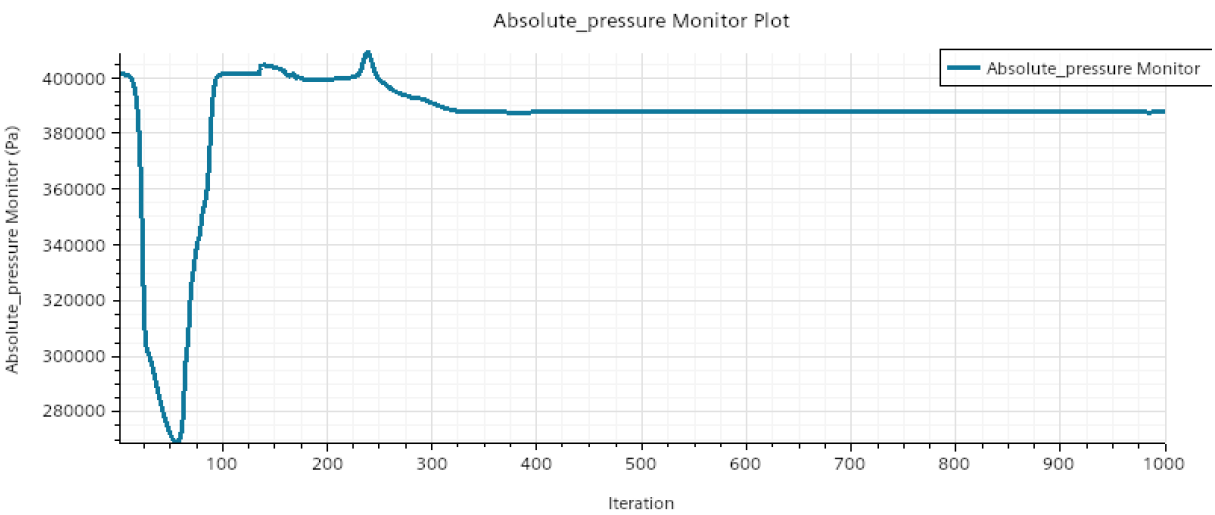


Fig. 15. Absolute pressure (300kPa)

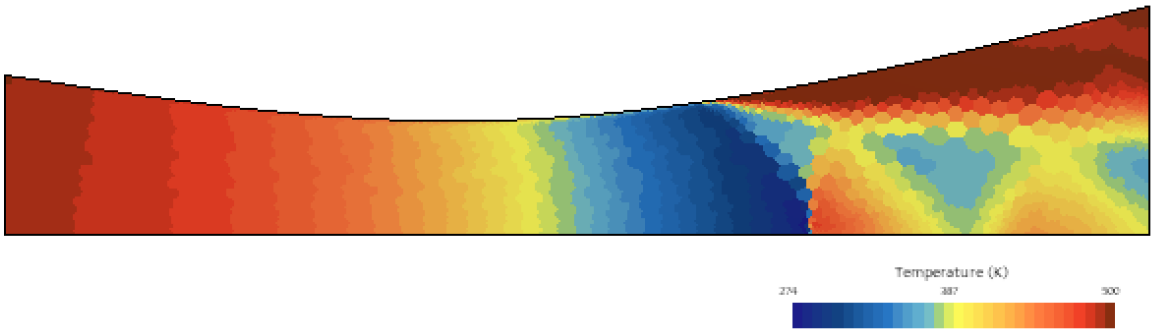


Fig. 16. Temperature (300kPa)

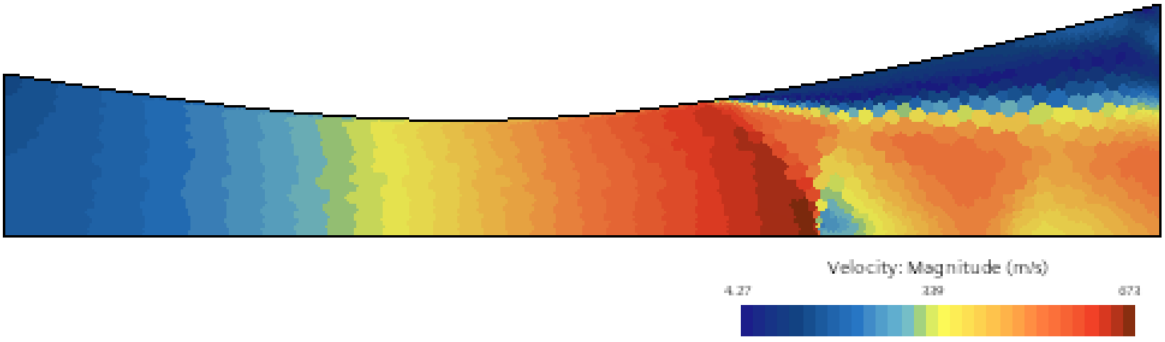


Fig. 17. Velocity magnitude (300kPa)

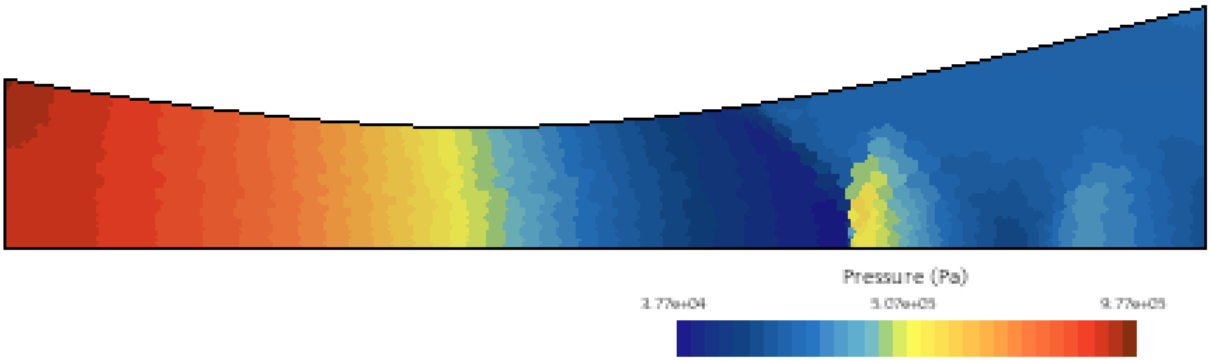


Fig. 18. Pressure (300kPa)

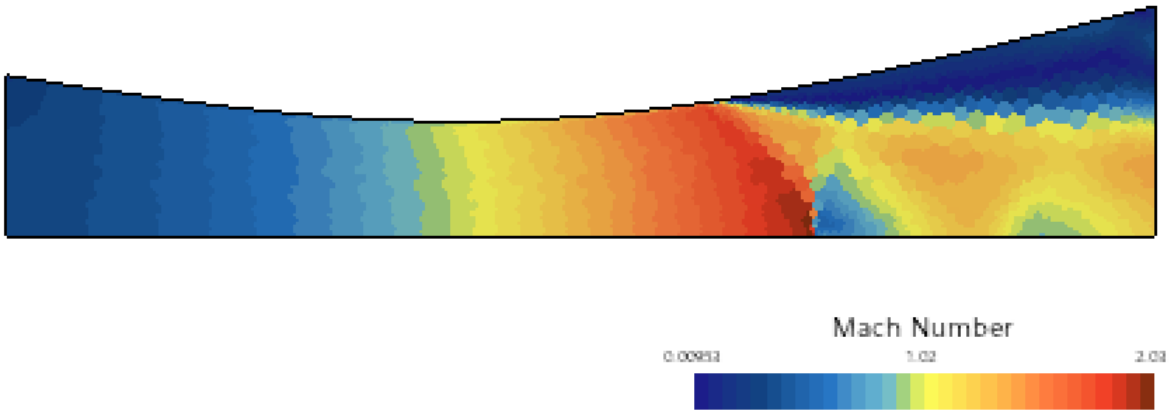


Fig. 19. Mach number (300kPa)

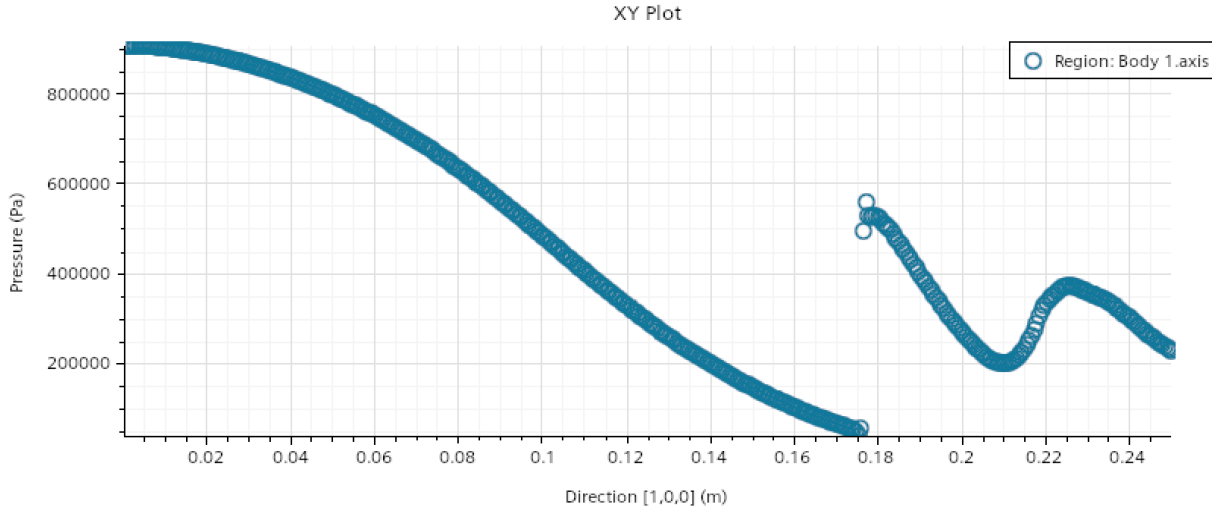


Fig. 20. XY Plot : pressure (300kPa)

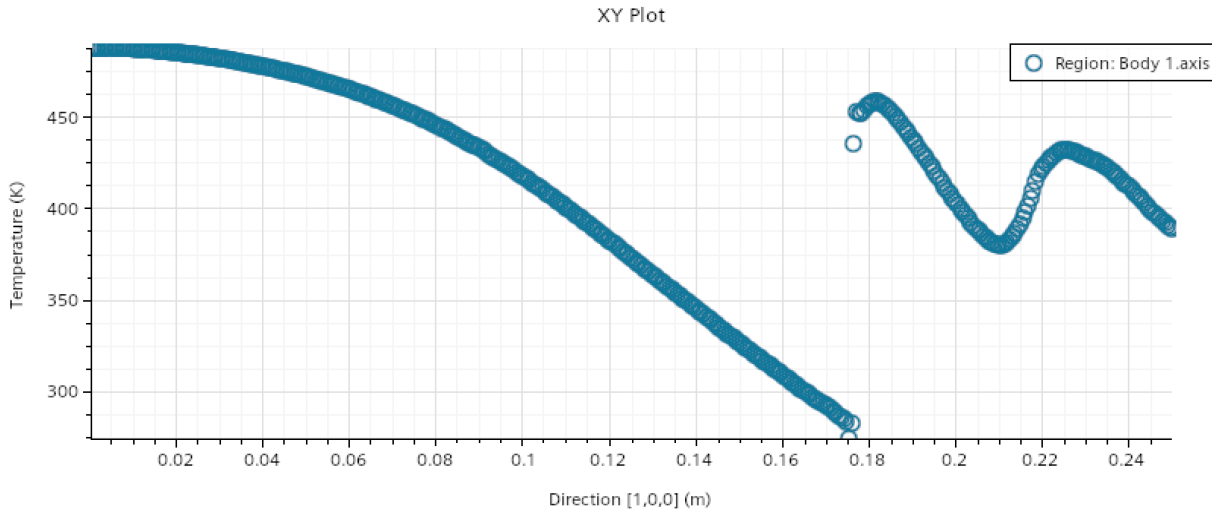


Fig. 21. XY Plot : temperature (300kPa)

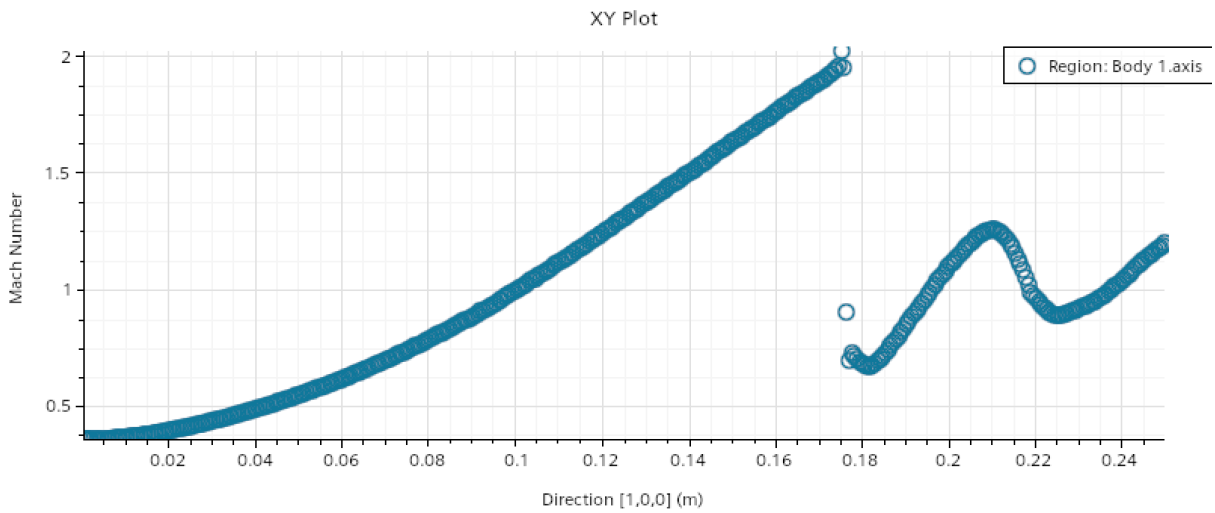


Fig. 22. XY Plot : mach number (300kPa)

4.3.3. **Scalars and XY Plots (for 500kPa)**

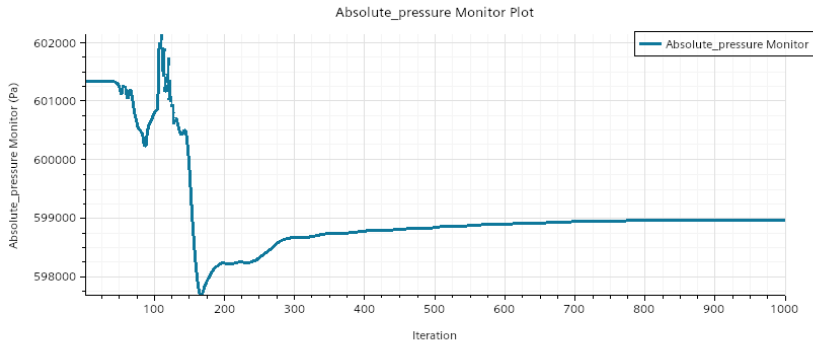


Fig. 23. Absolute pressure (500kPa)

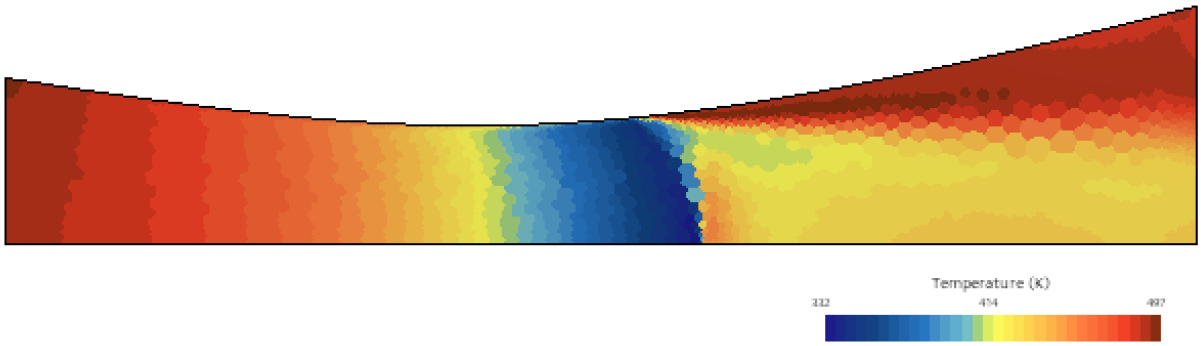


Fig. 24. Temperature (500kPa)

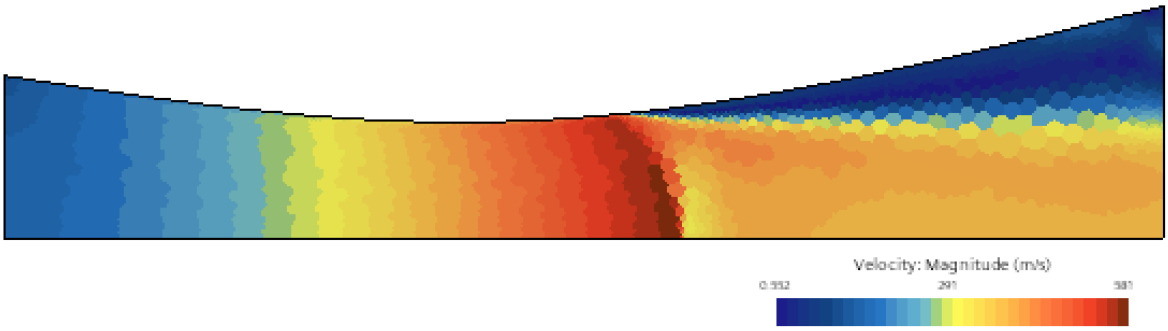


Fig. 25. Velocity magnitude (500kPa)

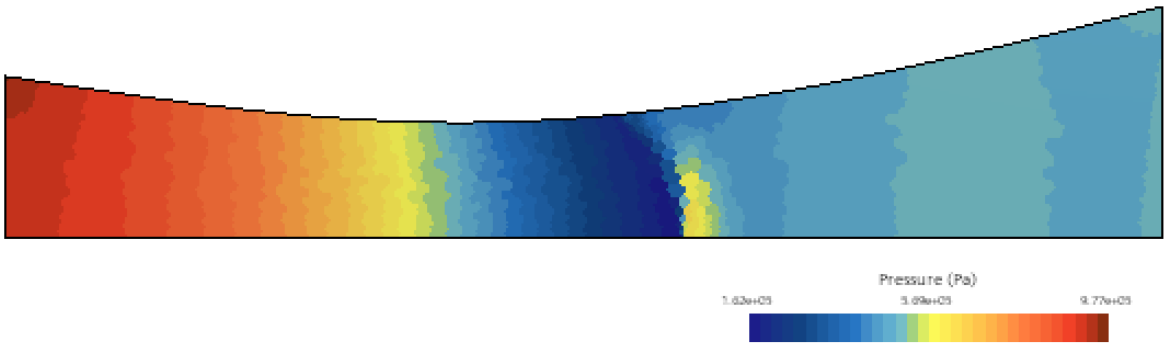


Fig. 26. Pressure (500kPa)

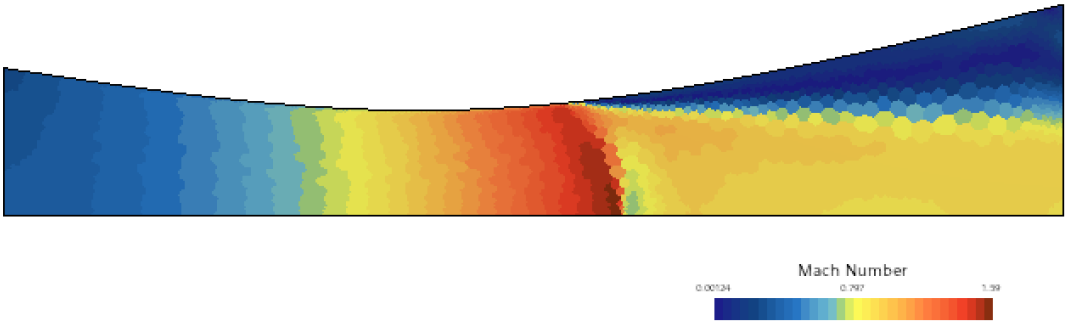


Fig. 27. Mach number (500kPa)

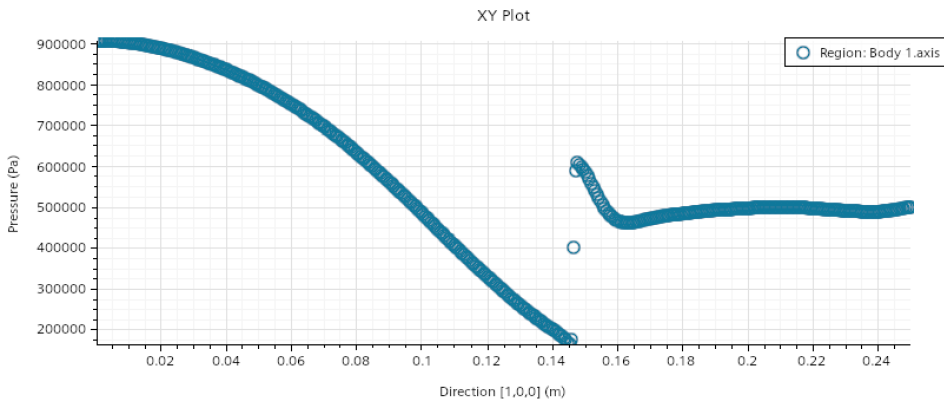


Fig. 28. XY Plot : pressure (500kPa)

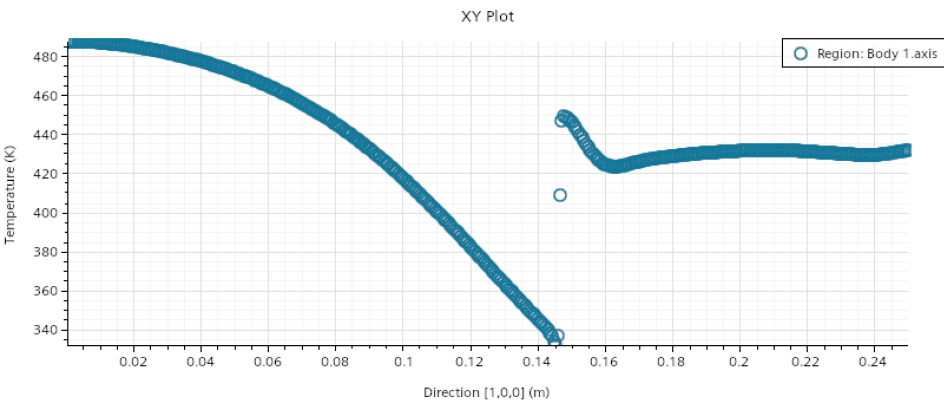


Fig. 29. XY Plot : temperature (500kPa)

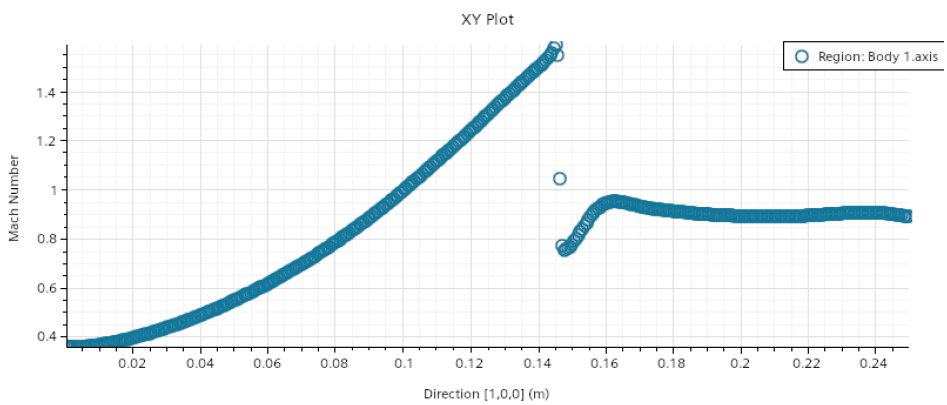


Fig. 30. XY Plot : mach number (500kPa)

5. Comparative Analysis: Influence of Outlet Pressure

The performance of a Laval nozzle is governed by the pressure ratio between the inlet (P_0 = 1000 kPa) and the outlet (P_s). This section analyzes the flow behavior at 29795.0138 Pa, 300 and kPa 500 kPa.

5.1. Flow Regime and Shock Wave Phenomenon

When the outlet pressure is increased to 300 or 500 kPa, the nozzle operates in a highly **over-expanded regime**. Because P_s is much higher than the pressure the flow would naturally reach at the exit, the supersonic flow is forced to adapt through a **normal shock wave** inside the divergent section.

5.1.1. Scalar Scene Observations

As illustrated in Fig. 25, 26, and 28, the shock wave is clearly visible as a sharp, high-gradient transition zone near $x \approx 0.145$ m:

- **Mach Number (Fig. 28):** The flow accelerates to $Ma \approx 1.6$ before the shock, then instantly drops to subsonics levels ($Ma < 1$), marked by the transition from red to blue regions.
- **Temperature (Fig. 25):** There is a brutal heating effect. The static temperature, which was dropping due to acceleration, suddenly rises to nearly 450 K as kinetic energy is dissipated into internal energy.

5.1.2. Numerical Discontinuities in XY Plots

The XY plots along the axis (Fig. 29, 30, 31) provide a mathematical proof of the shock wave at $x \approx 0.145$ m:

- **Pressure Jump (Fig. 29):** The static pressure exhibits a vertical discontinuity, jumping from $\approx 200,000$ Pa to $\approx 600,000$ Pa. This is the hallmark of the conservation of momentum across a shock.
- **Velocity Collapse (Fig. 31):** The Mach number sags from a supersonic peak (1.6) down to 0.8. After this point, the flow is subsonic.

5.2. Theoretical Validation and Assumptions

This behavior perfectly illustrates the core principles of fluid mechanics:

- **Continuity Equation:** Despite the shock, mass is conserved. The nozzle remains "choked" at the throat ($Ma = 1$), keeping the mass flow rate constant.
- **Effect of Section Variation (dS):** According to the Hugoniot relation, in a divergent section ($dS > 0$), a subsonic flow ($Ma < 1$) must decelerate. This is observed after the shock ($x > 0.15$ m), where the Mach number continues to decrease slightly as the pressure recovers.
- **Bernoulli and Energy:** While Bernoulli's equation is valid along streamlines for perfect fluids, the total pressure decreases across the shock due to entropy production (energy dissipation by friction/viscous effects within the shock layer).

5.3. Convergence Analysis

The absolute pressure monitor (Fig. 24) shows that the solution reaches a stable state at approximately 599,000 Pa after 600 iterations. The initial oscillations (0 – 200 iterations) represent the numerical movement of the shock wave until it finds its equilibrium position where the internal pressure matches the imposed back-pressure of 500 kPa.

5.4. Synthesis: 300 kPa vs 500 kPa

Compared to the 300 kPa case, the shock wave at 500 kPa has moved **upstream** (closer to the throat). This confirms that as the back-pressure increases, the shock propagates further into the nozzle to allow for a larger pressure recovery zone in the subsonic divergent section.

6. Requested Work

This section summarizes the compliance of the report with the initial requirements and provides the specific justifications requested.

1. **Mesh Scene (0.6 mm):** The baseline mesh used for the analysis features a base size of 0.6 mm as requested. The mesh topology, including the refined boundary layers, is illustrated in **Figure 4** (Page 9).
2. **Justification of Physical Models:**
   - **Steady/Coupled Flow:** Necessary for high-Mach compressible flows where pressure and velocity are strongly linked.
   - **Ideal Gas:** Required to account for density variations (ρ) as a function of temperature and pressure, essential in the divergent section.
   - **2D Axisymmetric:** Chosen because the nozzle geometry is a surface of revolution; this reduces computational cost while maintaining full 3D flow physics.
   - $k - \omega$ **SST:** A robust turbulence model that accurately handles the strong pressure gradients and potential flow separation after a shock wave.
3. **Flow Behavior Analysis:** The scalar scenes for the design condition, 300 kPa, and 500 kPa cases provide a visual synthesis of the Mach number acceleration and the shock wave location. See **Figures 8 to 11** and **Figures 25 to 28**.
4. **Pressure Profile along Nozzle Axis:** The pressure evolution for all three tested cases is plotted and interpreted in the comparative section. The numerical plots (e.g., **Figure 12** and **Figure 29**) confirm the isentropic expansion followed by the pressure jump across the shock.
5. **Temperature and Mach Number Profiles:** These profiles were extracted along the centerline (axis) and are presented in **Figures 22, 23, 30 and 31**. They highlight the conversion of thermal energy into kinetic energy.
6. **Mass Flow Rate Comparison:** The analytical mass flow rate was calculated as $\dot{m}_{th} \approx 3.61$ kg/s. As shown in the Mass Flow Monitor (**Figure 15**), the numerical result converges to approximately 3.9 kg/s. The small discrepancy (approx. 8%) is attributed to the inclusion of viscous effects (boundary layer displacement thickness) in the CFD model, which are neglected in the 1D isentropic theory.
7. **Comparison of P_s = 300 kPa and P_s = 500 kPa:** This detailed comparison is the core of our analysis in **Section 4.4**. It demonstrates that increasing the back-pressure forces the shock wave to move upstream (closer to the throat) to allow for sufficient pressure recovery.

7. Conclusion

This study validated the isentropic relations for a Laval nozzle. The numerical results for mass flow rate and exit pressure showed strong agreement with analytical theory, confirming the accuracy of the StarCCM+ coupled solver for compressible flows.

References

Kdous, W., & Abassi, W. (2026). *TP2: Modeling a Laval Nozzle*. IPSA.